A Multispectral Remote Sensing Crop Segmentation Method Based on Segment Anything Model Using Multistage Adaptation Fine-Tuning

Binbin Song, Hui Yang, Yanlan Wu, Peng Zhang, Biao Wang, *Member, IEEE*, and Guichao Han

***Abstract*— Multispectral information is crucial for remote sensing crop monitoring, but current methods struggle with inadequate feature extraction, leading to poor generalization and incomplete segmentation. The segment anything model (SAM) shows significant potential for generalization across fields, offering a promising solution for crop monitoring. This article introduces a crop segmentation method based on SAM using multistage adaptation fine-tuning, namely MAF-SAM, effectively utilizing information from multispectral remote sensing and the transfer and generalization abilities of SAM. In its first stage, MAF-SAM employs a prefix adapter to extract primary low-level multispectral features and compresses them into three channels to meet the requirements of subsequent stages. The second stage introduces a low-rank adaptation (LoRA) fine-tuning strategy to inject crop-specific knowledge into the image encoder, enhancing MAF-SAM's adaptability in particular crop segmentation tasks. In the third stage, it utilizes a mask decoder with no-prompt embedding to automatically generate masks with accurate class information. MAF-SAM achieves $F1$ scores and Intersection over Union (IoU) for soybean and corn of 0.9294, 0.8680, 0.8760, and 0.7723, respectively, along with a Kappa coefficient of 0.9543. It demonstrates superior temporal and spatial transfer capabilities relative to five other advanced segmentation methods in our study area.**

***Index Terms*— Adaption, crop segmentation, fine-tuning, high-resolution multispectral images, segment anything model.**

I. Introduction

DEEP learning methods have been widely applied to various vision fields, demonstrating exceptional performance in applications such as medical diagnosis [1], video surveillance [2], surface defect detection [3], and remote sensing building change detection [4]. However, in the field of multispectral remote sensing crop segmentation, certain challenges still restrain the performance of deep learning methods, waiting for additional in-depth exploration for more precise monitoring and identification of crops, thereby contributing to food security [5].

Different from traditional visual images with only the red, green, and blue bands, agricultural monitoring frequently employs multispectral images, which possess additional bands like near-infrared [6], [7] and short-wave infrared [8], [9]. The information derived from these multiple bands, including invisible spectral ranges, is crucial for discerning land cover, especially for different crops [10]. Furthermore, it is not easy to generate pixel-level large-scale crop samples due to the complexity of crop representation in remote sensing images, obvious variations in crop shapes and structures, high homogeneity with other vegetation, significant differences in growth cycles, and the lack of coherent time-series data [11]. Given the difficulty in mining complex information of multispectral and the limited availability of sample datasets, deep learning models encounter the problem of insufficient learning of multispectral features of crops, thereby constraining the generalization ability and accuracy of the model.

Single pixel-based deep learning methods were proposed in the early stages. Utilizing all bands of multispectral information [12] as primary features, in conjunction with hand-crafted statistical features of vegetation indexes (VIs) [13], such as NDVI [14], [15], GNDVI [16], NDRI [17], and other VIs [18], [19] as inputs for CNN or RNN models [20], [21], [22], this approach has yielded excellent results on MODIS and other low-resolution satellite images. Nonetheless, as Earth satellite observation technology advances, high-resolution satellite data has become the primary data source for remote sensing in crop monitoring [23]. With improved resolutions of images, like PlanetScope data at 3 m resolution, GF-1 data at 2 m resolution, and UAV data at 0.1 m resolution, different crops in images present the phenomenon that intraclass differences increased, while the interclass differences decreased. While the single pixel-based extraction methods struggle to fully take into account the diversity of samples and sense the contextual information [24], the extraction results have serious salt-and-pepper phenomena and obvious fragmentation.

In contrast to the single-pixel inversion method, the semantic segmentation approach is capable of learning complex crop

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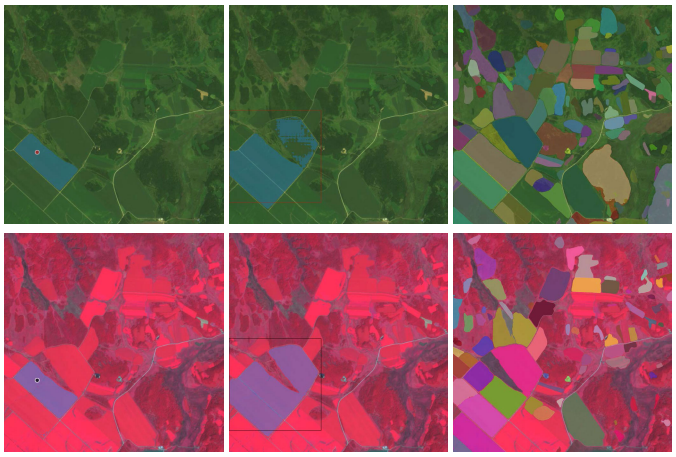

Fig. 1. Segmentation results obtained by SAM with different prompt modes. The first and second lines show the segmentation results of the combination of true and false color bands, respectively. In both scenarios, neither point, box prompt nor automatic mode is capable of automatically segmenting the entire image with class information. Crops exhibit greater reflectance in the near-infrared band, making them more distinguishable in the false color combination. (a) Point prompt. (b) Box prompt. (c) Automatic mode.

characteristics from texture, shape, color, and other aspects. This makes it more suitable for high-resolution remote sensing crop segmentation tasks and enables more precise and rapid agricultural mapping [25], [26], [27], [28], [29]. Nevertheless, the huge imaging variation among crops and the scarcity of pixel-level samples constrain the adequacy of different crops' feature learning of multispectral images, which continues to limit the generalization capability and accuracy of models.

The segmentation anything model [30] is the first large vision model proposed. It uses a large amount of training data to build a model with a large number of parameters through supervised learning. This model exhibits the strong transfer and generalization abilities to efficiently perform tasks like instance and semantic segmentation, object detection, and more [31], [32], [33], in various downstream applications without new training. It shows that segment anything model (SAM) can capture high-level semantic information of images more accurately compared with other deep learning segmentation methods, which are trained with fewer samples and fewer parameters. Therefore, the emergence of SAM introduces innovative ideas to improve the generalization ability of the crop extraction method and the precision of segmentation results for remote sensing under realistic scenarios.

However, similar to other state-of-the-art (SOTA) methods [34], [35], [36], [37], [38], SAM is primarily developed for three-band natural images. Although suitable band combinations can be chosen as input to directly apply to SAM like true or false color combinations, this approach fails to utilize all bands simultaneously and mine more important information beneficial for crop extraction. As shown in Fig. 1, which illustrates the segmentation results of two band combinations using different prompt modes, respectively. In addition, while interactive segmentation methods using point prompts or box prompts can generate relatively favorable local masks, they fall short of addressing the requirements of large-scale automatic segmentation mapping. The three segmentation modes are all incapable of automatically segmenting the entire image with accurate class information masks simultaneously. Therefore, it still needs further exploration to adapt SAM to multispectral remote sensing crop segmentation and fully utilize its robust generalization capabilities.

Based on the above analysis, this article proposes a crop segmentation method based on segment anything model using multistage adaptation fine-tuning, to fully exploit the generalization capability of SAM and extend its application from three-band images to multispectral remote sensing images. This method mainly consists of three fine-tuning adaptation stages. In the first adaptation stage, a prefix adapter is designed to extract potentially vital information from all spectral bands and compress principal features and semantic information into three channels. This compressed channel information will suit the requirement of the SAM's image encoder, initially adapt to the remote sensing crop segmentation task, and then be passed to the low-rank adaptation (LoRA) fine-tuning stage. In the next stage, LoRA layers are embedded into the image encoder through the low-rank decomposition, promoting the learning and extraction abilities of high-level semantic information about specific crops, thus further improving the adaptability for the crop segmentation task. In the end postfix adaptation stage, the no-prompt embedding mode is utilized for automatic whole-image segmentation and generates correct class information guided by a loss function. Finally, a series of comparative and ablation experiments in the study area are conducted to comprehensively validate the spatial and temporal transfer capabilities of SAM using multistage adaptation fine-tuning (MAF-SAM), along with the efficacy of its components and strategies. It demonstrates that the proposed MAF-SAM is suited for multispectral remote sensing crop segmentation tasks and can deliver superior segmentation results with precise class information.

In summary, the main contributions of this work are presented as follows.

1) A novel fine-tuning method using multistage to gradually guide SAM to adapt to the precise multispectral remote sensing crop segmentation task.
2) A prefix adapter is proposed to extract the primary features and semantic information of multibands and compress it into three channels by mainly using large kernel depthwise separable convolutions. It initially adapts the proposed model to multispectral remote sensing images and extends the applicable range of SAM from three-band images to multiband images.
3) A LoRA, based on rank decomposition, is proposed to inject crop-specific information into the image encoder of SAM. Since only a limited number of trainable parameters are updated, it greatly preserves the transfer and generalization capabilities of SAM and further augments its adaptability for crop segmentation tasks.

II. Related Works

In this section, we briefly review related work about deep learning methods for remote sensing crop segmentation and some fine-tuning strategies for SAM.

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A. Deep Learning Methods for Remote Sensing Crop Segmentation

Following the successful application of deep learning semantic segmentation methods in computer vision, their advanced capabilities have been transferred to agricultural remote sensing monitoring [26], [27], [28], [29]. DCNN-based semantic segmentation networks for remote sensing crop recognition in agricultural monitoring include fully convolutional networks [39], upsampling pyramid networks [40], [41], and multiscale feature fusion networks [42], [43]. In the early stage, the segmentation model in the field of computer vision is directly applied to remote sensing images. For instance, Zhao et al. [44] applied the Unet method on three-band multispectral UAV imagery for assessing rice lodging areas. Du et al. [43] applied DeeplabV3+ to worldview-2 images, consisting only of RGB bands, for crop and noncrop identification segmentation. Considering the challenge in extracting context information in high-resolution remote sensing images, Wang et al. [45] introduced CCTNet, coupling CNN, and Transform, which combines CNN and Transform. This method effectively combined the local detail extraction capabilities of CNN with the global information extraction of Transform, presenting promising performance on a three-band UAV crop dataset. Furthermore, Xu et al. [42] proposed MP-Net, employing a multilayer pyramid module to aggregate global crop information continuously in remote sensing images. Recognizing the unique noise characteristics in remote sensing images, Gerry proposed ODCNN-CCM [46], featuring adaptive Wiener filters for image preprocessing and dolphin swarm optimization (DSO) coupled with a deep stacked denoising autoencoder (DSDAE). Considering the characteristics of multispectral images and fully exploiting the channel information in multispectral images, the segmentation method is improved in the data input of the model. Wang et al. [40] took the 13 bands of Sentinel 2 data as input to Unet++ for the segmentation of crops like corn, peanut, and rice. In addition, improvements can be made in promoting the correlation of information between multispectral channels. Bian et al. [47] introduced a channel attention mechanism based on Unet, thereby enhancing the model's spectral feature extraction capabilities. However, these segmentation models are always limited by the number of samples and parameters, and cannot fully mine internal important information of crops in multispectral remote sensing [25].

B. Fine-Tuning Strategies

Despite SAM exhibiting its impressive zero-shot generalization capability, it cannot directly offer correct class information for various target crops owing to the absence of specialized data. Consequently, the utilization of SAM to guide downstream segmentation tasks from unknown knowledge has become a significant area of research. Current research tends to focus on refining the fine-tuning strategy. Some efficient parameter fine-tuning methods previously utilized to fine-tune large models in natural language processing (NLP) [48], [49], [50] offer valuable reference for SAM fine-tuning in downstream segmentation tasks. These include adapter-based, selective parameter fine-tuning, prompt-driven fine-tuning, and LoRA. The Adapter fine-tuning method involves freezing the weights of each layer of the image encoder and incrementally training the weights of the bypass branches. Furthermore, retraining the prompter encoder or mask decoder augments the segmentation network with specific knowledge or visual cues pertinent to downstream data. For instance, Ding et al. [51] developed a convolutional adapter, primarily composed of convolutional layers, to fine-tune FastSAM for remote sensing building change detection. Chen et al. [52] and Liu et al. [53] constructed an MLP adapter, based on MLP layers, to fine-tune SAM for object detection and audiovisual segmentation (AVS), respectively. Liu et al. [54] engineered a Pooling adapter, utilizing the pooling layer, to fine-tune SAM for shield tunnel leakage segmentation. A simple adapter method, namely Conv-LoRA [55], which integrates Convolutional with LoRA layers, has been introduced for medical image segmentation. However, the application of these methods to crop segmentation in actual multispectral remote sensing scenarios remains unexplored.

III. Method

In this section, we first introduce the overall model architecture of the proposed MAF-SAM for multispectral remote sensing crop segmentation. Subsequently, we introduce the three adaptation stages, respectively.

A. Overall Architecture of MAF-SAM

Fig. 2 illustrates the overall architecture of MAF-SAM. It consists of three adaptation stages, namely prefix, LoRA, and postfix. Spectral data from the five bands of either GF-1 or PlanetScope serve as the input for MAF-SAM. To make full utilization of information from multiple bands, in the first adaptation stage, the prefix adapter compresses the dimensionality of high-dimensional multispectral data into three channels, making it initially apt for crop segmentation tasks. This adaptation meets the input requirements of SAM's image encoder and simplifies the difficulty in the next fine-tuning stage. Subsequently, the compressed primary crop features and low-level semantic information are passed into the second adaptation stage. Within the frozen image encoder, the LoRA fine-tuning layers are embedded into each transformer block and concentrate on acquiring high-level semantic features about crops, thereby further enhancing MAF-SAM's adaptability for crop segmentation tasks. Then, these high-level semantic features are finally passed to the third adaptation stage to execute mask output adaptation. At this postfix adaptation stage, the mask decoder receives no-prompt embedding to automatically generate masks of the entire image. The segmentation head of the mask decoder has been modified to enable the output of multiclass crop segmentation masks guided by the loss function.

B. Prefix Adapter

The prefix adapter is designed to extract essential low-level semantic information from multiband images and compress the

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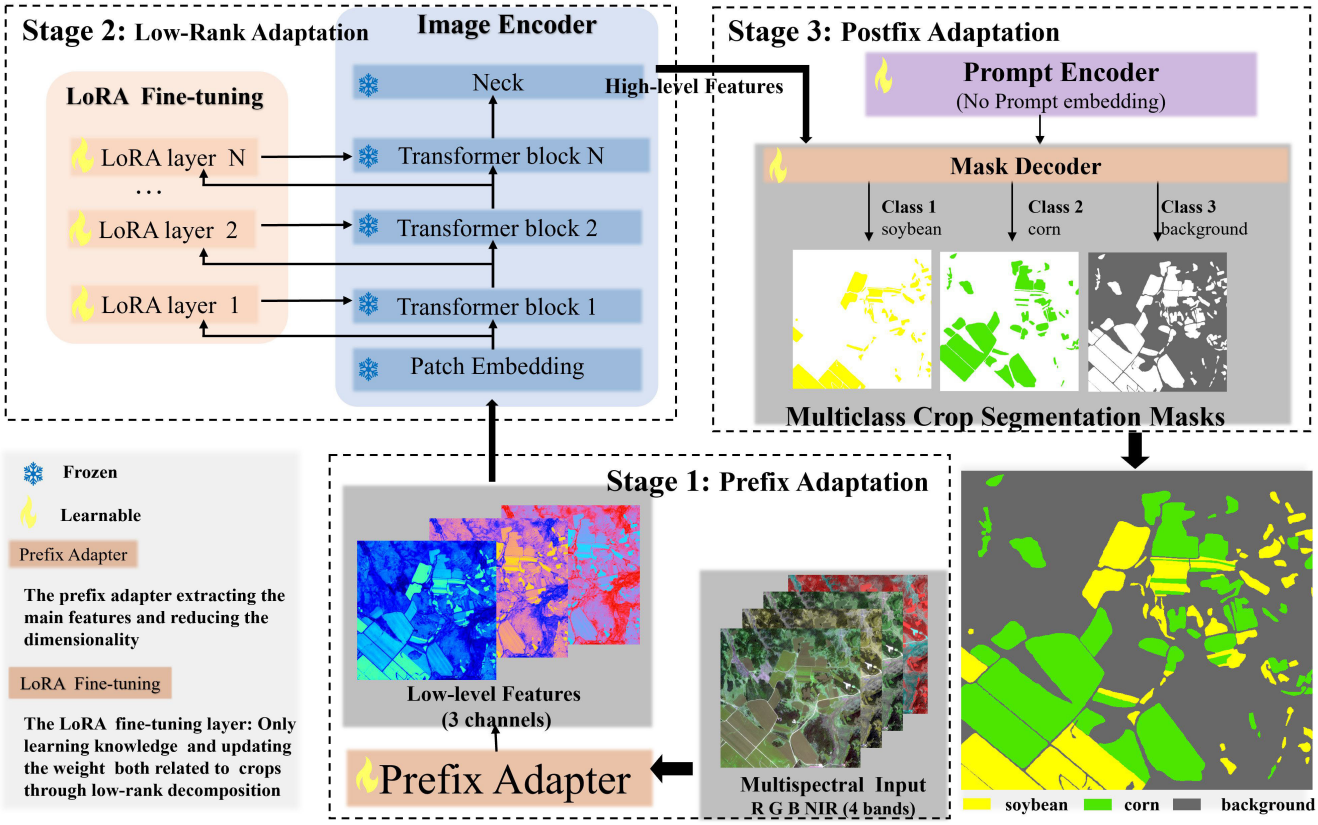

Fig. 2. Overview of the proposed MAF-SAM. The adaptability to crop segmentation is enhanced gradually, utilizing distinct adaptation methods across its three stages. In these stages, MAF-SAM achieves effective dimensionality reduction of multispectral data, fine-tuned learning of crop-specific knowledge, and the generation of multiclass segmentation masks in turn. The weights of the prefix adapter, LoRA layers, prompt encoder, and mask decoder are learnable, contrasting with the frozen weights of the image encoder.

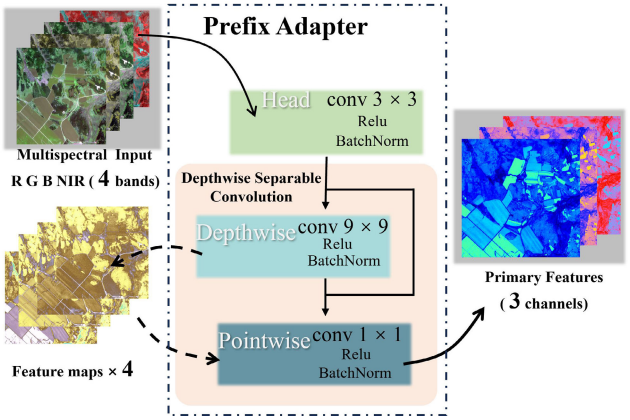

Fig. 3. Overview of the proposed prefix adapter. It extracts low-level semantic information from each channel of the multispectral data and performs channel fusion and dimensionality reduction, thereby guiding the subsequent LoRA fine-tuning stage in learning high-level semantic features of crops.

dimensions to three channels while preserving the fundamental features of multiband images. As mentioned above, the vital crop characteristic information hidden in the invisible spectrum is essential for remote sensing crop extraction. Although SAM exhibits advanced feature capture and generalization capabilities, its image encoder is limited to processing only three channels of input. Despite it is possible to replicate or select a combination of three bands, like the RGB band combination in Sentinel-2 or the VH, HH, and HV polarization combination in Sentinel-1, to meet data entry specifications [56]. Nevertheless, this limitation hinders SAM from directly utilizing the spectral information of multiple bands. Consequently, the prefix adaptation was developed, and its architecture is illustrated in Fig. 3, primarily consisting of depthwise separable convolution. The prefix adapter consists of a 3×3 head convolution, a 9×9 depthwise convolution, and a 1×1 pointwise convolution. Each convolution operation is followed by ReLU activation and batch normalization layers. The outputs of the 3×3 and 9×9 convolution are concatenated with the final 1×1 convolution through residual style, which aims to prevent vanishing gradients. The 9×9 depthwise convolutions efficiently extract long-range spatial information and retain the primary low-level semantic features of each channel. The 1×1 pointwise convolution promotes the fusion of band information. Large-scale convolution kernels are better for capturing global information, yet decrease performance and speed [57]. In contrast, depthwise separable convolution balance feature extraction, model parameters, and computational cost [58]. Following the prefix adapter, the model initially completes the adaptation of multispectral remote sensing data, supplying extensive low-level semantic information that guides the image encoder in learning high-level semantic information about crops in the next stage.

C. Fine-Tuning Strategy of Low-Rank Adaptation

The rank of a matrix denotes the number of linearly independent row or column vectors it contains. In machine

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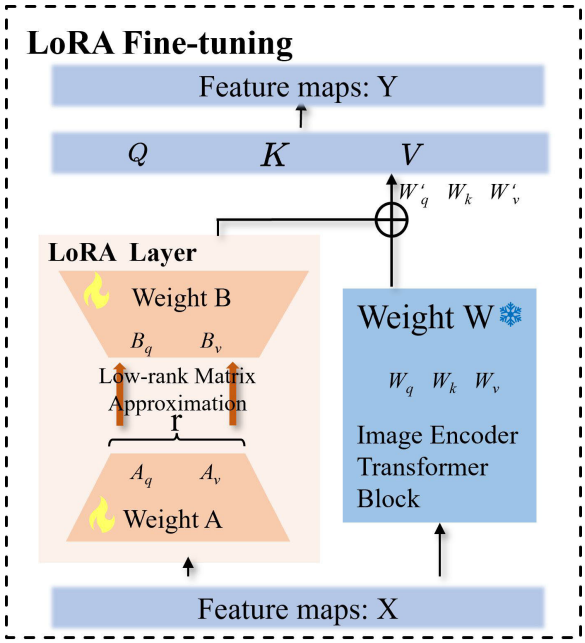

Fig. 4. Overview of the LoRA fine-tuning strategy. The LoRA layer is embedded into each transformer block of the image encoder. W_q, W_v, and W_k are the projections of W, respectively. A_q, A_v, B_q, and B_v are the learnable parameter weights in the LoRA layer.

learning, matrices often represent fully connected layers, which are typically overparameterized. This suggests that significant features can be decomposed by computing the matrix's rank, enabling the representation of high-dimensional features with a low-rank matrix. Features extracted from remote sensing images by SAM are often excessively complex. Conversely, the intrinsic dimensional features relevant to crop segmentation tasks are much smaller [59]. Moreover, the large number of parameters in SAM makes them unsuitable for extensive fine-tuning. The LoRA [60] strategy presents an innovative approach to utilize large vision models, initially developed for the NLP domain. This strategy allows SAM to update only a small subset of model parameters during the fine-tuning process, significantly reducing the computational resources needed for the crop segmentation task. Additionally, this method provides increased training throughput and decreased inference delay, all while maintaining the model's segmentation performance. The LoRA fine-tuning strategy in MAF-SAM is illustrated in Fig. 4. Three types of weights are involved: the weight $W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ for each image encoder transformer block, and the low-rank weight matrices $A \in \mathbb{R}^{d_{\text{in}} \times r}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$. Specifically, the weight W is frozen to ensure that the prediction results of the image encoder derived from SAM are not changed before fine-tuning. Additionally, A is randomly initialized using the Kaming method, and B is initialized to zero. During the beginning phase of fine-tuning adaptation, the high-dimensional features extracted by the image encoder transformer layer are initially projected into three downward components: Q (query), K (key), and V (value). Subsequently, the two weight matrices A and B in the LoRA layer consisting of fully connected layers will complete the adaptive adjustment of projections Q and V and make minor updates to the weights through low-rank decomposition.

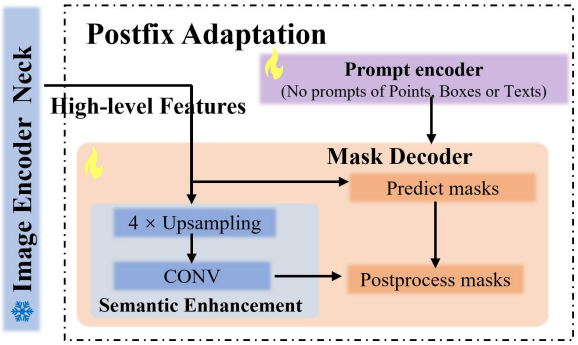

Fig. 5. Overview of the postfix adaptation. The prompter encoder and mask decoder are inherited from SAM. The no-prompted default embeddings and high-level semantic features from the LoRA fine-tuning stage are jointly fed into the mask decoder. A semantic enhancement branch is added to the mask decoder to improve the representation of the semantics of crops.

The LoRA fine-tuning process can be described as follows:

$$\begin{aligned} Y &= XW' \\ &= XW + XAB \end{aligned} \tag{1}$$

where $W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ is the weight of a transformer block in the image encoder, d_{in} is the input token's dimension, and d_{out} is the output token's dimension. $X \in \mathbb{R}^{b \times s \times d_{\text{in}}}$ and $Y \in \mathbb{R}^{b \times s \times d_{\text{out}}}$ are the tokenized high-level feature maps, respectively. b is the batch size and s is the tokens' length. r is the rank size of $A \in \mathbb{R}^{d_{\text{in}} \times r}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$, where $r \ll \min\{d_{\text{in}}, d_{\text{out}}\}$.

From this, we can obtain the update process in fine-tuning the weight W'

$$\begin{aligned} W' &= W + \Delta W \\ &= W + AB \end{aligned} \tag{2}$$

where $W' \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ merges the value of BA into W. Extending to the transformer architecture, this formula can be readily applied to the Q, K, and V matrices generated by the multihead self-attention mechanism for X. However, to conserve computational resources, our focus will be on the weight modifications influenced by Q and V

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{\text{out}}}}\right)V \tag{3}$$

$$\begin{aligned} Q &= XW_q' \\ &= XW_q + XA_qB_q \end{aligned} \tag{4}$$

$$\begin{aligned} V &= XW_v' \\ &= XW_v + XA_vB_v \end{aligned} \tag{5}$$

$$K = XW_k \tag{6}$$

where W_q, W_v, and W_k are the projections of W, whose parameter weights are fixed and cannot be changed. A_q, A_v, B_q, and B_v are the learnable parameter weights in the LoRA layer.

D. Postfix Adaptation

A principal objective of this article is to produce multicategory segmentation masks with high precision. Fig. 5 displays the structure of the postfix adaptation, which consists of the learnable prompter encoder and mask decoder derived from

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SAM and the semantic enhancement (SE) branch. It receives no-prompt default embeddings and high-level features from the image encoder. Although SAM can achieve precise object segmentation at specific locations by using certain point or box prompts. However, the mask decoder in SAM struggles to segment the entire image while simultaneously generating precise boundary and category information. Consequently, we eliminate the specific prompt engineering and only utilize the default no-prompt embedding to generate masks of the entire image. Furthermore, the segmentation head of the mask decoder is modified to enable the output of multiclass crop segmentation masks, guided by the loss function. These two modifications compel updates in the weights of the prompt encoder and mask decoder, thereby leading to their parameters being trainable. Additionally, the SE branch is incorporated into the mask decoder to improve the accuracy of boundaries in segmentation masks by promoting the representation of high-level features. Within this branch, the neck feature output from the image encoder will be upsampled four times in size and directly applied to multicategory segmentation output. This output is then integrated with the mask decoder's low-resolution segmentation result and interpolated sampling to match the original image's resolution.

IV. Materials

In this section, we provide a comprehensive description of our study area, the utilized multispectral remote sensing images, the methodology for producing and dividing the crop segmentation dataset, and the associated data processing method.

A. Study Area

Farms located in the northeast region of Hulun Buir, Inner Mongolia Autonomous Region of China constituted the selected study area. This area includes Ganhe Farm (*A*) and Ganhe Farm (*B*) in the Molidawadawoerzu Autonomous Banner, Dongfanghong Farm (*C*), Yili Farm (*D*) and Guri Farm (*E*) in the Elunchun Autonomous Banner, and Dahewan Farm (*F*) in Zhalantun City. The types of crops, climate, soil types, and precipitation grown in these study subareas are generally similar, but differ to some extent, which can effectively test the applicability of the proposed method. Fig. 6 illustrates the different geographic of the six study farm regions, detailing the areas for sample data collection, testing, and the annotation of two crop types: soybean and corn. In these regions, remote sensing image data were collected to construct both a training and a testing dataset. Table I details the imaging time, image name, band information, and spatial resolution for each of the images employed in this study. In this study, a single temporal dataset was employed for crop type mapping due to the challenges in acquiring high-resolution remote sensing images with multiple temporal series and reducing the difficulty in creating the crop segmentation dataset.

B. Dataset Acquisition

Ganhe Farm (*A*), located between approximately 124°18′E to 124°54′E and 49°9′N to 49°22′N, a Chinese GF-1 satellite image at 2 m resolution in this region was utilized to create a training sample set for the deep learning segmentation model. This GF-1 satellite image was sourced from Hulun Buir State Farm. Utilizing visual interpretation with ground verification, 1031 soybean farmland plots and 775 corn farmland plots were meticulously labeled with complete boundary information. 669 training samples of 512 × 512 pixels were obtained through cropping finally. As shown in Fig. 6 and Table I, notable disparities exist in spatial resolution, imaging time, and imaging location between the images employed for training and testing. These variations are crucial in assessing the segmentation model's generalization capabilities, encompassing both spatial and temporal transfer abilities. These samples were used as training for the crop segmentation models.

In Ganhe Farm (*A*), nine Sentinel-2 image patches with different periods were used to verify the temporal transfer ability of the proposed model. Each image is 12 × 12 km in size. While in the remaining five research subregions—Ganhe Farm (*B*), Dongfanghong Farm (*C*), Yili Farm (*D*), Guli Farm (*E*), and Dahewan Farm (*F*)—five PlanetScope images were employed to verify the spatial transfer ability of the proposed model. For each image, two 5 × 5 km patches were randomly selected to evaluate the performance and accuracy of the segmentation method. These PlanetScope image patches cover a broad geographical range, with most located significantly distant from Ganhe Farm (*A*). The spatial disparity effectively validates the generalization of the proposed method. There are 19 evaluation patches in total, which corresponds to 160 images of 512 × 512 pixels, and the ratio of the training-to-testing set is about 6:1. Each PlanetScope satellite image is downloaded from the Planet website.

C. Data Preprocessing

The GF-1 image was transformed from a 16-bit to an 8-bit format with a series of preprocesses including radiometric calibration, atmospheric correction, panchromatic fusion, and geographic registration. As the PlanetScope images represent surface reflectance products sourced from the Planet website, additional preprocessing was not required. These images were similarly transformed from 16- to 8-bit format. Furthermore, the GF-1 image served as the reference for conducting relative radiometric corrections on all PlanetScope images. Five bands of Sentinel-2 image, including blue, green, red, and near-infrared, were selected, which were consistent with Gaofen-1 and PlanteScope images. Additionally, the images were converted from 16- to 8-bit. All the above-mentioned processing operations were performed by using the ENVI5.6 software.

V. Experiments and Results

This section first details the geographical distribution and imagery specifics of the crop segmentation test set, followed by an introduction to the qualitative evaluation metrics for the experimental results and the specifics of model training. Subsequently, the section presents both quantitative and qualitative findings from the comparative experiments, accompanied by an analytical assessment. Finally, the design of the ablation

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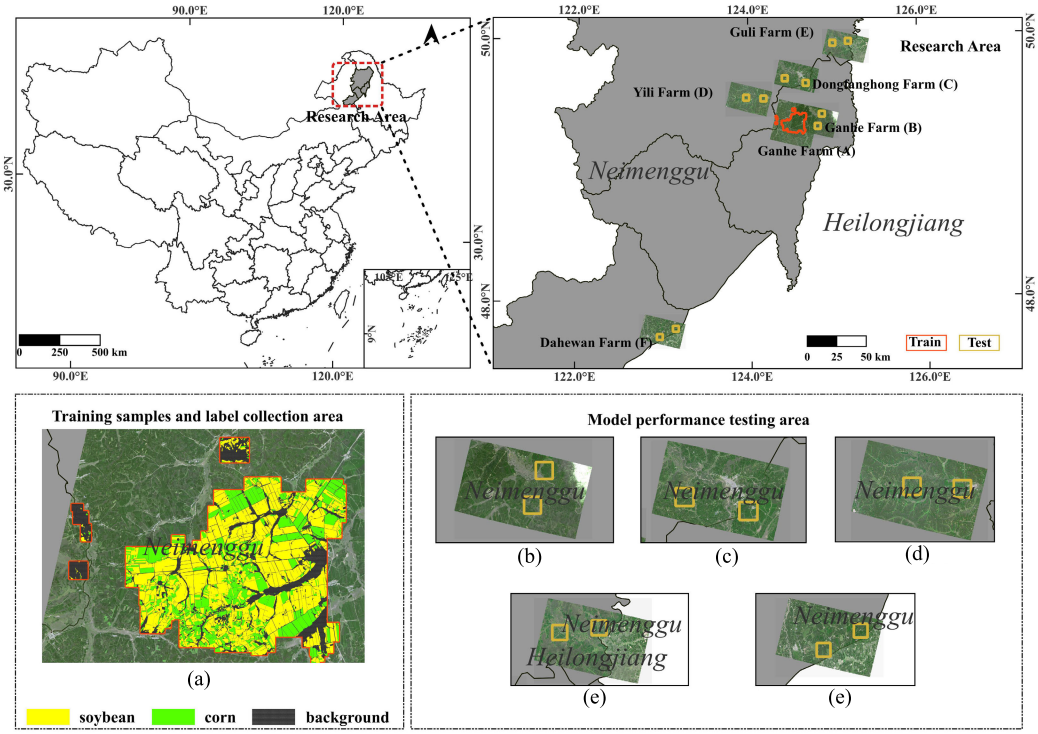

Fig. 6. Study area is in eastern Hulenber, Inner Mongolia, China.

TABLE I
FARM NAME AND ATTRIBUTE INFORMATION OF THE REMOTE SENSING IMAGES IN THE STUDY AREA

Dataset	Farm	Attribute Information				
		Imaging Time	Center Coordinates	Sensor	Resolution	Bands Used
Train	Ganhe Farm (A)	20220816	(E124.5, N49.4)	GF1_PMS1	2 m	Blue Green Red Nir-infrared
Test	Ganhe Farm (A)	20220622	(E124.5, N49.4)	Sentinel-2	10 m	
		20220707	(E124.5, N49.4)			
		20220720	(E124.5, N49.4)			
		20220801	(E124.5, N49.4)			
		20220806	(E124.5, N49.4)			
		20220809	(E124.5, N49.4)			
		20220816	(E124.5, N49.4)			
		20220821	(E124.5, N49.4)			
		20220831	(E124.5, N49.4)			
		20220910	(E124.5, N49.4)			
	Ganhe Farm (B)	20210818	(E124.8, N49.4)	PlanetScope	3 m	
	Dongfanghong Farm (C)	20220816	(E124.6, N49.7)			
	Yili Farm (D)	20220816	(E124.0, N49.6)			
	Guli Farm (E)	20220816	(E125.2, N49.9)			
	Dahewan Farm (F)	20220819	(E123.0, N47.8)			

studies is presented, along with both qualitative and quantitative outcomes and their corresponding analyses.

A. Test Dataset

The distribution of the test areas and the images used are detailed in the study area section, as illustrated in Fig. 6 and Table I, respectively. According to the data division in Table I, three types of comparative experiments were conducted. First, PlanetScope images in 2022 from different farms at the same growth stage were used to test the spatial transfer ability of the model: Dongfanghong Farm (C), Yili Farm (D), and Guri Farm (E). Then, Sentinel-2 image patches in 2022 of different growth cycles of crops in the Ganhe (A) farm were selected to verify the temporal transfer ability of the model. Additionally, the PlanetScope images of the Ganhe (B) region from 2021 were selected to verify the cross-year adaptation ability.

B. Evaluation Metrics

Accuracy, Recall, Precision, $F1$-score, Intersection over Union (IoU), and Kappa coefficient are standard metrics for evaluating remote sensing image segmentation. The IoU is

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commonly used to compare model performances, and the Kappa coefficient assesses the consistency between predictions and ground truth, serving as an important measure of remote sensing interpretation accuracy. In addition, the number of model parameters and GPU memory consumption are also important indicators to evaluate the performance of the model

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{7}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{10}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{11}$$

$$p_e = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{TN} + \text{FP})(\text{TN} + \text{FN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})^2} \tag{12}$$

$$\text{Kappa} = \frac{\text{Accuracy} - p_e}{1 - p_e} \tag{13}$$

where true positive (TP) means the number of instances where the model correctly identifies positive cases. True negative (TN) means the number of instances where the model correctly identifies negative cases. False positive (FP) means the number of nonobject pixels incorrectly classified as positive. False negative (FN) means the number of object pixels incorrectly classified as negative. p_e estimates the likelihood that any agreement between the observed accuracy (actual correct predictions) and the expected accuracy (predictions by chance) is just coincident, which is crucial for Kappa statistics.

C. Implementation Details

All experiments were performed using the "vit_b" version of SAM, implemented on the PyTorch 1.12.1 framework with a single NVIDIA 3090Ti-24G Tensor Core GPU, an eight-core CPU, and 64 GB of RAM. We employed the widely used segmentation loss function, Dice loss, along with an AdamW optimizer at a learning rate of 1e-3 to train all models. Dice loss is often used in semantic segmentation tasks to improve the segmentation performance and attenuate the effects of imbalance class problems. The number of batch sizes and epochs are set to 4 and 60, respectively. Additionally, we utilized ReduceLROnPlateau to optimize the learning rate. All experiments across five compared methods utilizing the same training, validation, and testing datasets

$$\text{DiceLoss} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{14}$$

where $|X \cap Y|$ is the intersection between the predicted label $|X|$ and the actual label $|Y|$. The $|X|$ and $|Y|$ subtables represent the number of elements of X and Y.

D. Comparison Results

Over the past decade, a variety of deep-learning methods have been developed for remote sensing crop segmentation. Representative segmentation methods include but are not limited to, convolutional neural networks (CNNs) [61], Contextual deep CNN [62], FCN [39], Unet [40], CENN [63], Unet++ [41], PSPnet [64], SegNet [65], DeeplabV3+ [43], HyFormer [66], and CTFuseNet [67]. To demonstrate the superiority of MAF-SAM, it is compared with five other typical and novel methods: Unet [40], PSPnet [64], ST-Unet [38], CPCPU [47], and CMFnet [68]. Figs. 7 and 8 show the segmentation results on the ten test regions in terms of temporal and spatial transfer abilities, respectively. The first three columns present the farm name, images, and crop ground survey results of the test area, respectively, while columns 4–8 provide visual comparisons of the different methods. The yellow-labeled areas represent soybean, green-labeled areas denote corn, and black areas indicate nonsoybean and noncorn background objects.

1) Spatial Transfer Ability: Fig. 7 displays the segmentation results for Dongfanghong Farm (C), Yili Farm (D), Guli Farm (E), and Dahewan Farm (F), respectively. The images from these different regions are captured in mid-August 2022 and align with the period of the training images but with different planting structures. Fig. 7 reveals the segmentation results of all methods are relatively accurate. However, in several regions, MAF-SAM demonstrates superior performance over other models, like Yili ($D1$), Yili ($D2$), Dahewan ($F1$), and Dahewan ($F2$), especially for corn. From Fig. 7, it can be observed that the segmentation results of all methods are relatively accurate. In regions with complex and varied terrain, such as Yili Farm and Dahewan Farm, MAF-SAM exhibits notably more accurate segmentation visually compared to the other five contrastive methods. It effectively removes the background and accurately distinguishes between corn and soybean, with fewer segmentation errors. This suggests that MAF-SAM can adapt to farms with different cropping structures and possesses spatial transfer capabilities. The qualitative evaluation indices for the five farms are presented in Table II. MAF-SAM maintains the highest scores in $F1$ and IoU for both soybean and corn, achieving 0.8827 and 0.9403, and 0.7948 and 0.8894, respectively. Additionally, the Kappa coefficient also reached its peak at 0.9635. These metrics further confirm the superior spatial transfer capability of MAF-SAM.

2) Temporal Transfer Ability: Fig. 8 illustrates the NDVI trends of soybean and maize at different growth stages from sowing to harvest in Ganhe Farm (A), along with Sentinel-2 images at various phenological stages. Soybean and corn at various phenological stages exhibit different morphology, texture, and spectral characteristics, beneficial for testing the model's generalization ability. Fig. 9 presents the segmentation results of MAF-SAM and five other comparison methods at different phenological stages. From the visual point of view, MAF-SAM achieves the most accurate segmentation of soybeans and corn, followed by ST-Unet. However, the soybean and corn segmented by Unet, PSPnet, and CACPUnet are significantly incomplete and cannot distinguish between soybean and corn effectively. CMCFnet, on the other hand, failed to segment soybean and corn in Sentinel-2 images. Exploring the cause of this situation, it may be because the

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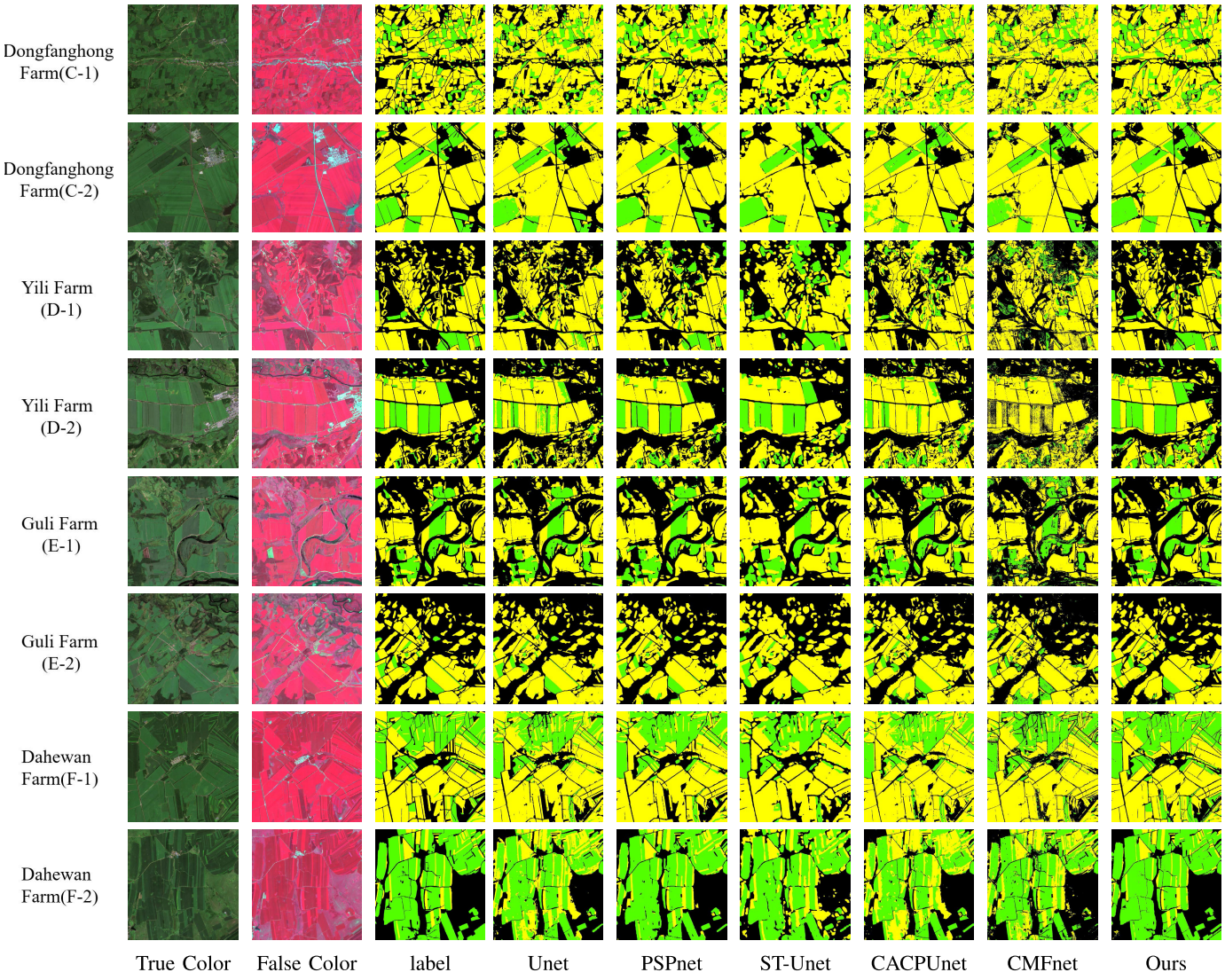

Fig. 7. Qualitative results between MAF-SAM and five other typical deep learning models on the four farms spanning different long distances, which belong to the same growth cycle in 2022. Eight PlanetScope image patches are used for segmentation to evaluate the spatial transfer ability of the model. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

TABLE II

QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON SPATIAL TRANSFER ABILITY IN DONGFANGHONG FARM (C), YILI FARM (D), GULI FARM (E), AND DAHEWAN FARM (F). THE BEST RESULT IS BOLD

Methods	Accuracy		Precision		Recall		F1		IoU		Kappa
	soybean	corn	soybean	corn	soybean	corn	soybean	corn	soybean	corn	
Unet	0.7975	0.9984	**0.9249**	0.8075	0.7656	**0.9730**	0.8300	0.8682	0.7297	0.7884	0.8031
PSPnet	0.9285	0.9897	0.8273	0.8820	0.8916	0.9522	0.8533	0.9146	0.7556	0.8455	0.9254
ST-Unet	0.9872	9446	0.8035	0.8042	0.9414	8908	0.8592	0.8286	0.7658	0.7233	0.9158
CPCPUnet	0.6922	**0.9990**	0.7589	0.7622	0.6759	0.9716	0.7023	0.8366	0.5731	0.7437	0.7054
CMF-net	**0.9979**	0.6665	0.7698	0.6518	0.9329	0.5661	0.8324	0.5914	0.7284	0.4884	0.6733
MAF-SAM	0.9743	0.9857	0.8315	**0.9215**	**0.9465**	0.9605	**0.8827**	**0.9403**	**0.7948**	**0.8894**	**0.9635**

sensors of the Sentinel-2 satellite are quite different from those of PlanetScope or GF-1 satellite, including resolution, spectral characteristics and imaging time, which results in significant differences between the data feature distributions of soybean and corn and the training samples. Additionally, the scarcity of crop samples may lead to inadequate model training and insufficient learning of image features for soybean and corn. In a world, compared to MAF-SAM, the generalization ability of other comparison methods is significantly limited across different crop growth stages. In contrast, MAF-SAM effectively inherits the transfer and generalization capabilities of SAM. Fig. 10 presents the quantitative evaluation results of the segmentation performance of MAF-SAM and other comparison methods at different growth stages. MAF-SAM

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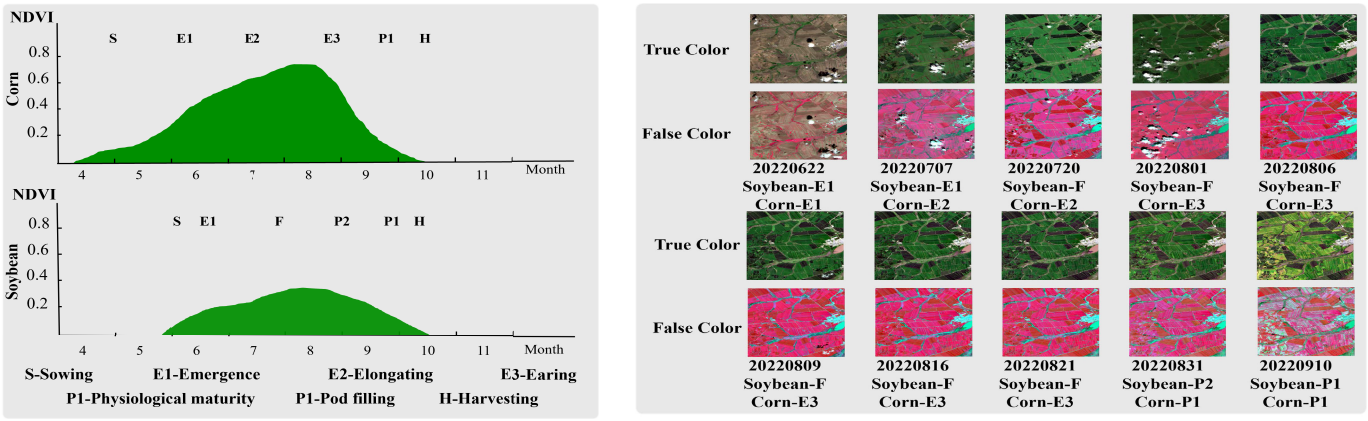

(a) (b)

Fig. 8. NDVI and remote sensing images of soybean and corn at different growth stages in Ganhe (A) farm. (a) NDVI curve of soybean and corns. The two crops each have six different growth stages. (b) Sentinel-2 time-series images of soybean and corn at different growth stages.

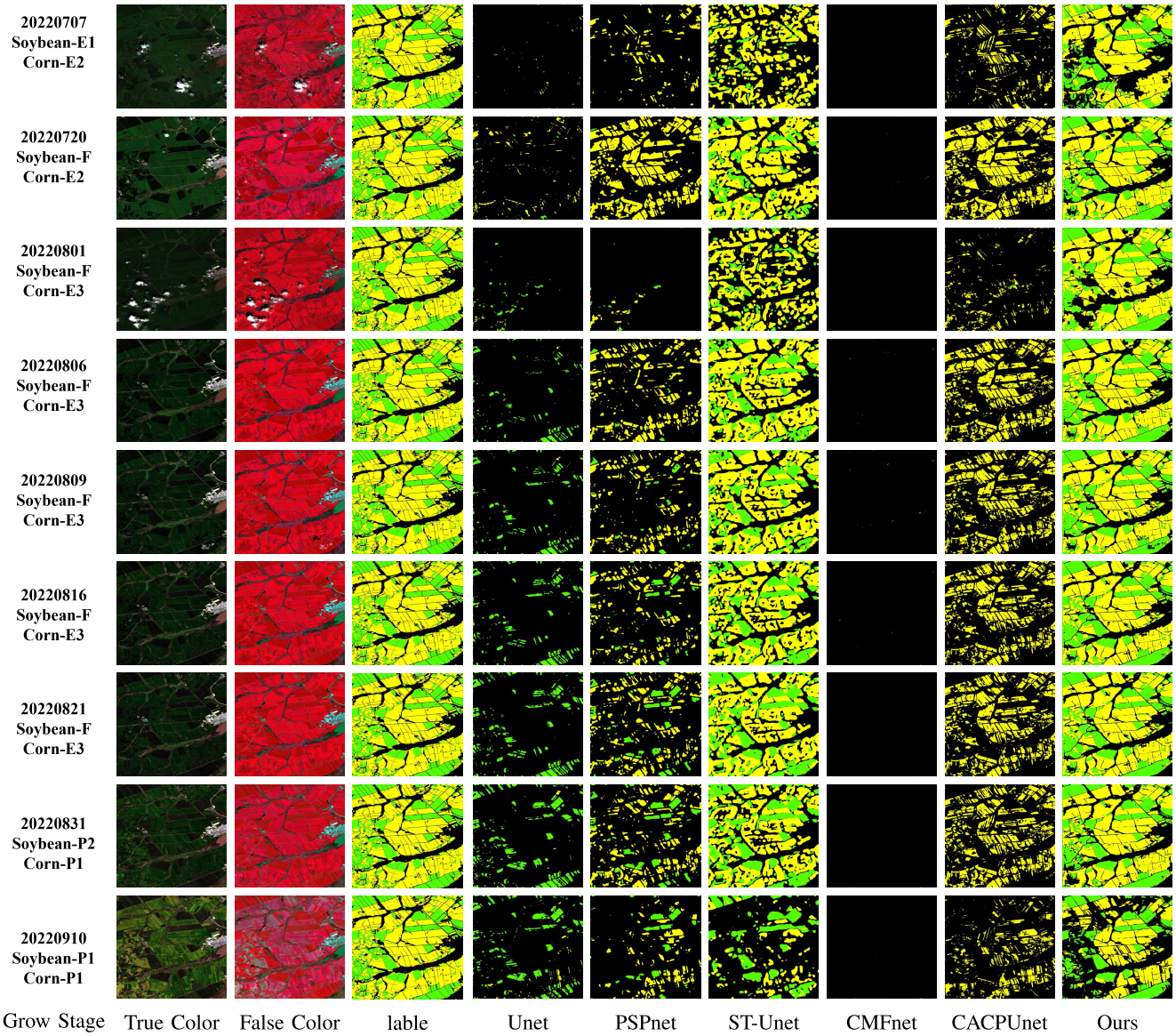

Fig. 9. Qualitative results between MAF-SAM and five other typical deep learning models at different crop growth stages on the Sentinel-2 images of Ganhe farm (A). $E1$, $E2$, $E3$, F, $P1$, and $P2$ mean Emergency, Elongating, Earing, Physiological maturity, and Pod filling, respectively. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

can maintain relatively high accuracy and stability of the segmentation in early, middle, and late crop growth stages.

3) Cross-Year Adopting Capability: The two test patches of the PlanetScope images from the Ganhe Farm (B) test

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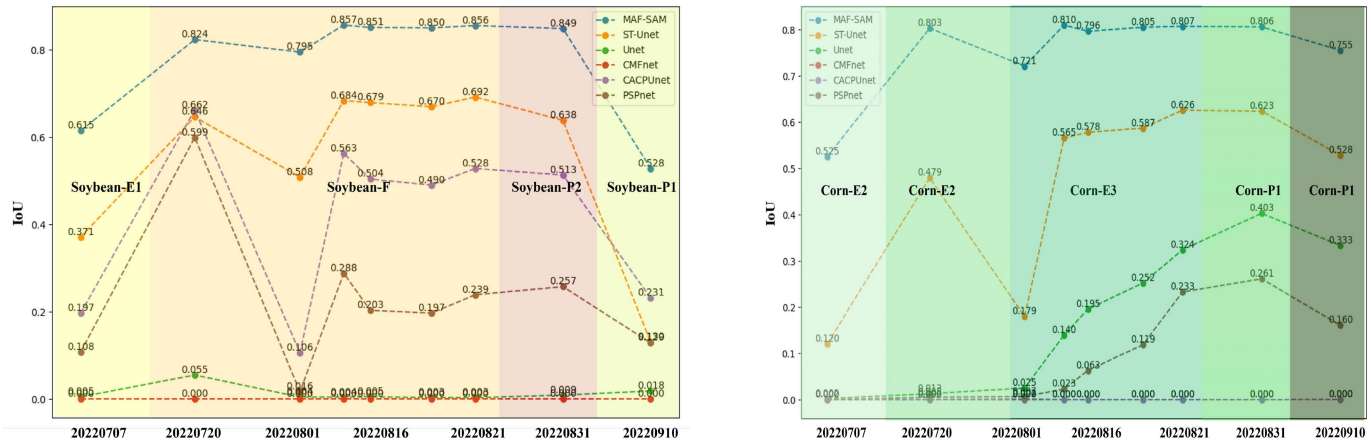

Fig. 10. Segmentation accuracy of each comparison method on Sentry 2 images at different growth stages in Ganhe (A) farm. (a) IoU of soybean in different growth stages. (b) IoU of Corn in different growth stages.

TABLE III

QUANTITATIVE RESULTS ON GANHE FARM (B). THE BEST RESULT IS BOLD

Methods	Accuracy		Precision		Recall		F1		IoU		Kappa
	soybean	corn	soybean	corn	soybean	corn	soybean	corn	soybean	corn	
Unet	**0.9998**	0.1765	0.4329	**0.7609**	0.9856	0.1744	0.6010	0.2779	0.4302	0.1633	0.1534
PSPnet	0.9966	0.1813	0.4484	0.4574	0.9727	0.1765	0.6128	0.2494	0.4431	0.1425	0.1596
ST-Unet	0.9997	0.3073	0.4294	0.6134	0.971	0.2974	0.5904	0.3975	0.4240	0.2491	0.28
CPCPU	0.9997	0.0702	0.3743	0.2187	**0.9935**	0.0700	0.5411	0.0874	0.3735	0.0457	0.0607
CMFnet	0.9993	0.1065	0.3891	0.3745	0.9805	0.0993	0.5561	0.1557	0.3865	0.0825	0.0923
MAF-SAM	0.9723	**0.9503**	**0.8340**	0.7223	0.9453	**0.9219**	**0.8857**	**0.8089**	**0.7952**	**0.6821**	**0.9171**

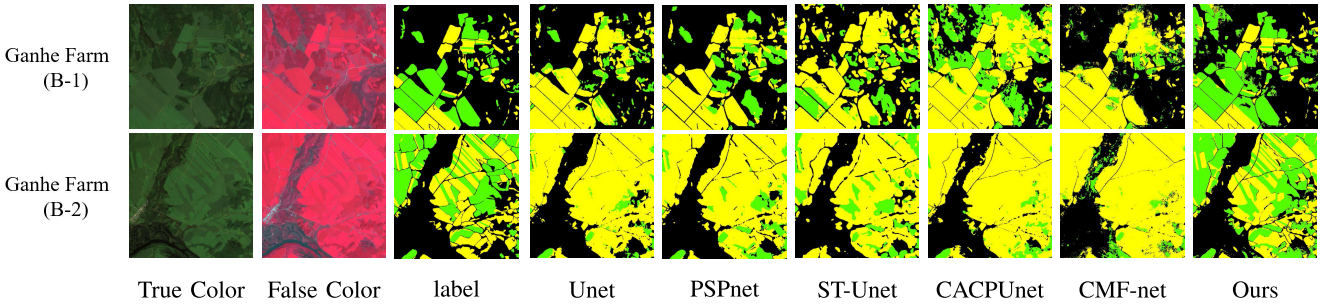

Fig. 11. Qualitative results between MAF-SAM and five other typical deep learning models on Ganhe Farm (B), an image belonging to the PlanetScope image taken photos on August 18, 2021, different from other images in 2022. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

area, shown in Fig. 11, were taken photo on August 18, 2021. Although these images share the same crop growth phase with the training data, there are notable differences in the time domain data distribution, presenting significant challenges to the model's cross-year adopting capability. It is visually obvious in Fig. 11 that the other five methods struggle to distinguish between the two crops in images taken in a different year. A large portion of corn is misclassified as soybean or background vegetation. In contrast, MAF-SAM is capable of identifying and segmenting soybean and corn crops more accurately. Table III presents the qualitative test results for the temporal transfer ability of each model in the Ganhe Farm (B). Corresponding to Fig. 11, due to the misclassification, the Accuracy and Recall for soybean of the other five models are disproportionately high, while those for corn were notably low. Additionally, the other five models show similar very low performance in $F1$, IoU, and Kappa. This phenomenon is frequently observed in the practical application of crop segmentation models in different years,

TABLE IV

MODEL PARAMETER AND COMPUTATIONAL RESOURCE COMPARISON

Methods	Parameters	Trainable parameters	GPU Memory
Unet	32.1 M	32.1 M	10.1 G
PSPnet	52.5 M	52.5 M	9.6 G
ST-Unet	30.92 M	30.92 M	5.6 G
CACPUnet	23.54 M	23.54 M	16.2 G
CMF-net	98.49 M	98.49 M	23.0 G
MAF-SAM	90.39M	3.96 M	7.6 G

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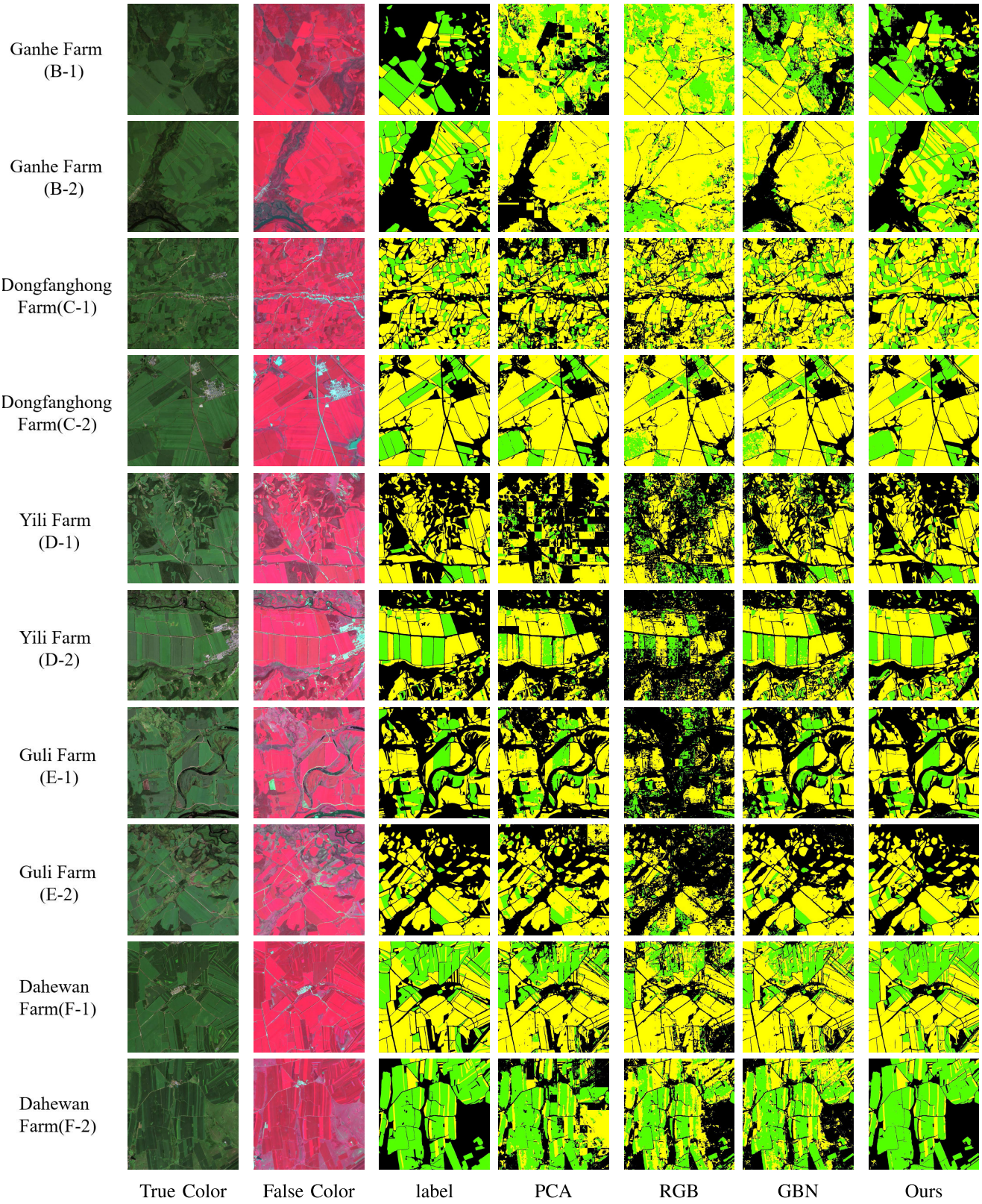

Fig. 12. Qualitative results of ablation experiments regarding the effect of not using the Prefix Adapter. RGB stands for true color band combination, GBN stands for false color band combination, and PCA stands for dimension reduction to three bands using PCA. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

as insufficient training samples and the complexity of crop phenotypes across different years result in inadequate learning of crop features. However, MAF-SAM achieved the highest scores in $F1$-score, IoU, and Kappa metrics for both corn and soybean, significantly surpassing the other five methods. Despite some degree of misclassification in its segmentation results, MAF-SAM still exhibits superior semantic segmentation capability and more stable time transfer ability compared

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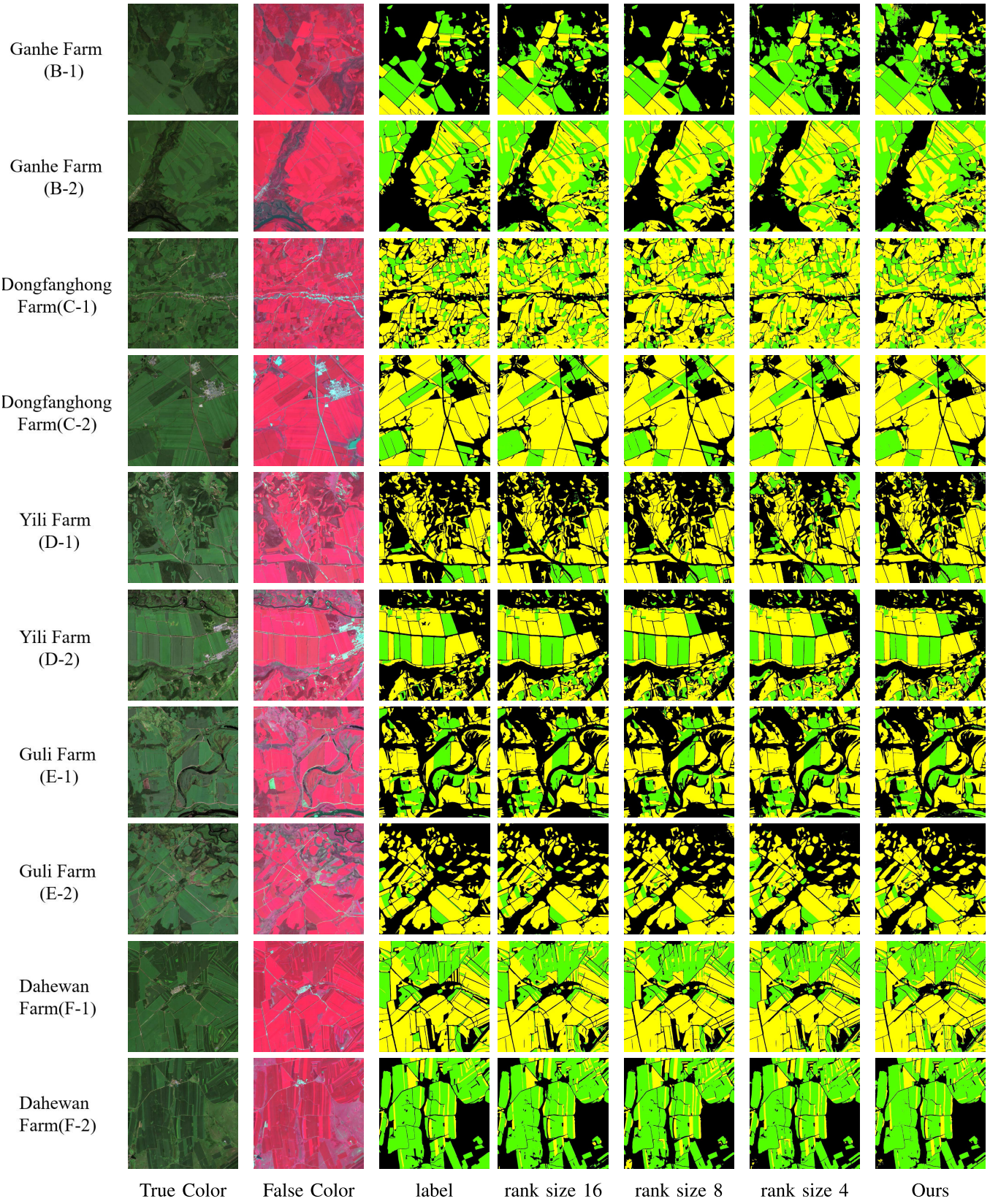

Fig. 13. Qualitative results of ablation experiments with different rank sizes in LoRA layers. The rank size in LoRA layers of MR-SAM is 1. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

to the other five methods. This shows that MAF-SAM has better cross-year adopting capability within the same growth period.

4) Parameters and Flops Computational Resource: Table IV displays the model parameters, trainable parameters, and computational resource consumption for various methods

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TABLE V

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE TEN PLANETSCOPE IMAGE PATCHES WITH DIFFERENT BAND COMBINATIONS OF INPUTS WITHOUT USING THE PREFIX ADAPTER. THE BEST RESULT IS BOLD

Methods	Accuracy		Precision		Recall		F1		IoU		Kappa
	corn	soybean	corn	soybean	corn	soybean	corn	soybean	corn	soybean	
RGB	0.6285	0.9943	0.6175	0.672	0.4294	0.7923	0.4850	0.6635	0.3513	0.5222	0.6226
GBNir	0.6489	**0.9983**	0.7435	0.7038	0.6025	**0.9622**	0.6472	0.7862	0.5177	0.6797	0.6570
PCA	0.6352	0.9620	0.6671	0.6592	0.5497	0.8054	0.5831	0.6988	0.4717	0.5858	0.5674
MAF-SAM	**0.9695**	0.9830	**0.8097**	**0.9040**	**0.9415**	0.9574	**0.8680**	**0.9294**	**0.7723**	**0.8706**	**0.9543**

TABLE VI

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE TEN PLANETSCOPE IMAGE PATCHES USING DIFFERENT RANK SIZES IN THE LORA LAYER. THE BEST RESULT IS BOLD

Rank size	Total trainable parameters	Accuracy		Precision		Recall		F1		IoU		Kappa
		corn	soybean	corn	soybean	corn	soybean	corn	soybean	corn	soybean	
16	4.51 M	0.9471	0.9833	0.8265	0.8825	0.9246	0.9459	0.8704	0.9110	0.7766	0.8412	0.9264
8	4.22 M	0.9256	**0.9933**	**0.9087**	0.8629	0.8867	0.9593	**0.8945**	0.9058	**0.8122**	0.8329	0.9146
4	4.08 M	0.9511	0.9911	0.7244	0.8746	0.9282	**0.9637**	0.8004	0.9150	0.6834	0.8470	0.9308
1 (MAF-SAM)	**3.96** M	**0.9695**	0.9830	0.8097	**0.9040**	**0.9415**	0.9574	0.8680	**0.9294**	0.7723	**0.8706**	**0.9543**

TABLE VII

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE PLANETSCOPE IMAGE PATCHES WITHOUT USING THE SEMANTIC ENHANCEMENT BRANCH. THE BEST RESULT IS BOLD

Methods	Accuracy		Precision		Recall		F1		IoU		Kappa
	corn	soybean	corn	soybean	corn	soybean	corn	soybean	corn	soybean	
No SE	0.8792	**0.9962**	**0.8909**	0.8274	0.8505	**0.9659**	0.8678	0.8826	**0.7816**	0.8033	0.8591
MAF-SAM	**0.9695**	0.9830	0.8097	**0.9040**	**0.9415**	0.9574	**0.8680**	**0.9294**	0.7723	**0.8706**	**0.9543**

when the batch size is 4. MAF-SAM inherits from SAM (version "VIT-b,") and Table IV indicates that the total number of model parameters is up to 90.34 M. However, the application of the LoRA fine-tuning strategy reduces the trainable parameter count to 3.96 M, significantly smaller than other comparison methods. Additionally, while the GPU memory consumption of MAF-SAM is not minimal, it remains smaller than that of classical segmentation models like Unet and PSPnet.

E. Ablation Study

In this section, ablation experiments are conducted on the same test regions to further demonstrate the influences of each component of the proposed method. Specifically, the experiments include: 1) *RGB:* the true color band input combination without using the prefix adaptation stage; 2) *GBNir:* the false color band input combination without using the prefix adaptation stage; 3) *PCA:* a three-band input combination obtained through principal component analysis (PCA); 4) *Rank Size:* employing different low-rank matrices in the LoRA fine-tuning strategy; and 5) *SE:* using the SE branch to improve the representation of high-level semantic information.

1) Effectiveness of the Prefix Adapter: To demonstrate the effectiveness of the prefix adapter, we conduct three sets of ablation experiments. In the absence of the prefix adapter, we utilize the RGB true color combination, the GBNir false-color band combination, and a three-band combination obtained through PCA, commonly employed in hyperspectral dimensionality reduction, to retrain and test the performance of the fine-tuned models. Fig. 12 and Table V present the segmentation results and qualitative accuracy metrics across ten test regions, respectively. The results indicate that using only true color or false color combinations is significantly less effective than band conversion through the prefix adapter. This suggests that the full information in the four-band combination is crucial for enhancing crop identification results. Additionally, the false color band combination's accuracy index is significantly higher than the true color band's, indicating the importance of the near-red band in remote sensing crop identification and segmentation. Furthermore, the PCA method's performance is not as effective as either the prefix adapter or the true color band combination. This may be attributed to PCA being a data preprocessing method whose dimension reduction results are not tailored to specific conditions in the training process and are influenced severely by the distribution of ground objects. The fact shows that it is effective to adapt multispectral data and fully utilize the multispectral information using the prefix adapter. Extracting low-level features promotes better guiding the fine-tuning of high-level semantic features.

2) Influences of the Size of Low-Rank Matrix in LoRA Fine-Tuning Adaptation: To ascertain the impact of the rank size in the LoRA fine-tuning strategy on model performance, we implement various low-rank matrices in our ablation experiments. Fig. 13 and Table VI display the qualitative and quantitative results of these experiments, respectively. The results indicate that increasing the rank size does not substantially improve the model's performance, and in some cases, it even leads to a significant decline in certain metrics. This may be explained that due to the adaptation of the prefix

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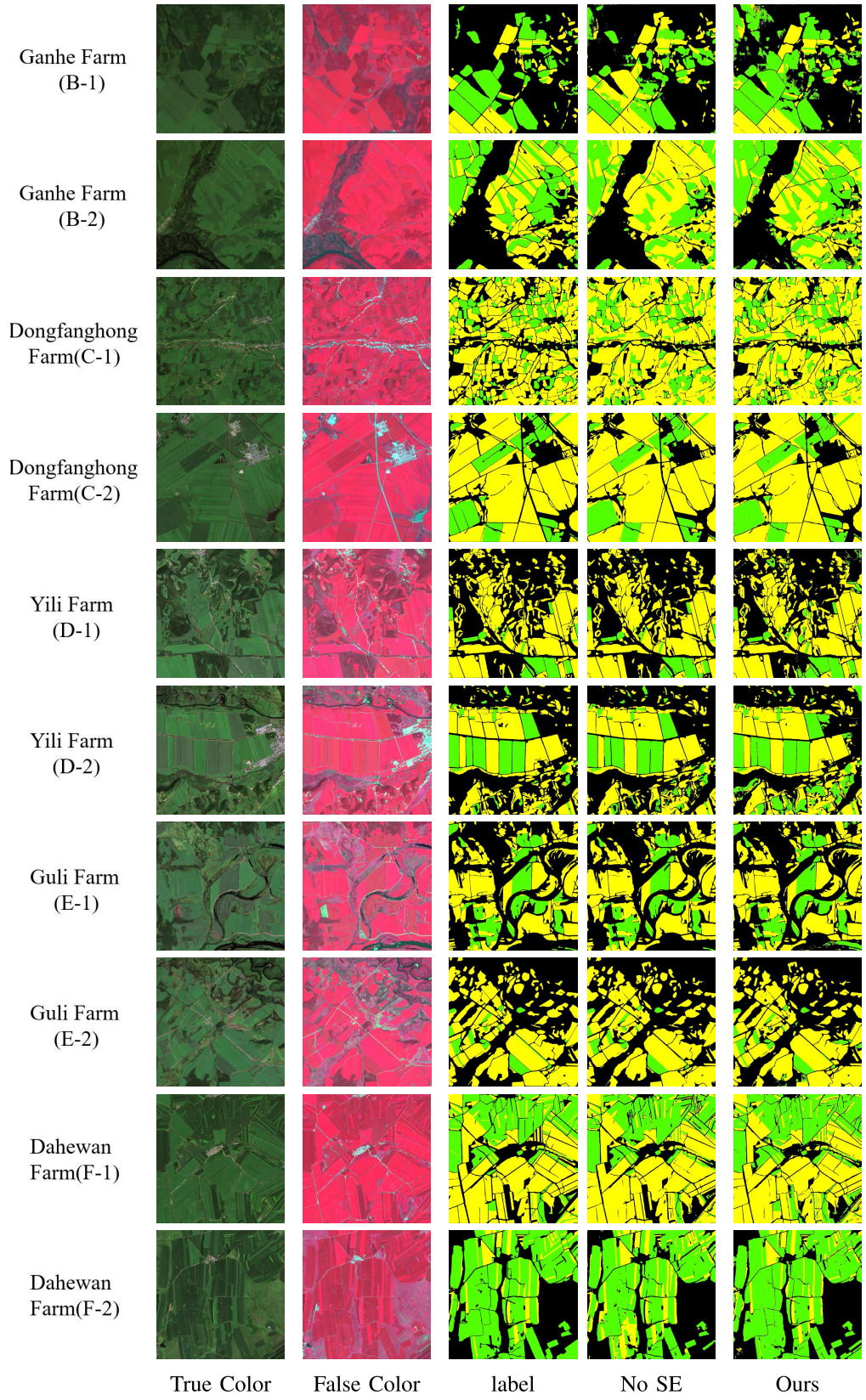

Fig. 14. Qualitative results of ablation experiments regarding the effect of without using the SE branch. The yellow labeled area represents soybean, the green labeled area represents corn, and the black area represents other nonsoybean and corn background objects.

adapter in the previous stage, which already extracts abundant low-level semantic information, resulting in a reduced need to update numerous parameters during the LoRA stage. Consequently, the rank-size of one suffices to convey the necessary intrinsic dimensional information in crops. A higher rank level introduces additional training parameters, potentially leading

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to redundant feature information and a decrease in model performance.

3) Effectiveness of the SE Branch in Mask Decoder: To assess the effectiveness of the SE branch in the mask decoder, an experiment is conducted without using the SE branch. Table VII and Fig. 14 show the qualitative and quantitative results of this experiment, respectively. When removing the SE branch, there was a slight improvement in the accuracy, recall, and IoU of soybean, as well as a marginal increase in the precision of corn. However, the average IoU, Kappa, and $F1$ scores for both soybean and corn decreased significantly, suggesting that the SE branch contributes to enhancing the presentation of high-level features thus improving the model's performance.

4) Discussion: Based on the qualitative and quantitative results, MAF-SAM effectively captures the foreground information of soybean and corn from remote sensing images, surpassing the other five methods. It also requires fewer fine-tuning parameters and less computational resource consumption. Although MAF-SAM's segmentation results are not entirely accurate, existing misclassification issues, particularly those involving background information, can potentially be mitigated by increasing additional background data or utilizing postprocessing algorithms to further enhance accuracy. Additionally, employing more advanced training methods such as other fine-tuning strategies or combined multiloss functions could potentially yield even higher performance in remote sensing crop segmentation.

VI. Conclusion

In this article, we propose a new method for multispectral remote sensing crop segmentation. Different from previous works, our method employs the large vision segmentation model with powerful transfer and generalization abilities to adapt multispectral data, which mainly utilizes multistage to gradually finish the fine-tuning process. The prefix adapter in the first adaptation stage compresses the dimensionality of high-dimensional multispectral data into three channels, making it initially apt for crop segmentation tasks. The low-level semantic information obtained in the prefix adaptation stage is used to guide the low-rank weight fine-tuning process of the image encoder in the next second stage. Finally, the accurate multiclass segmentation masks are generated in the third postfix adaptation stage through the fine-tuning process of no-prompt embedding. In diverse evaluation regions distant from the sample collection area, our proposed MAF-SAM achieves 0.8706, 0.7723, 0.9294, 0.8680, and 0.9543 in $F1$, IoU of soybean and corn, and overall Kappa mapping accuracy, respectively. Meanwhile, MAF-SAM shows strong temporal transfer ability for remote sensing crop segmentation in different growth stages and different years. Ablation studies further validate the effectiveness of these adaptations. MAF-SAM contributes a new method for using a large vision model for crop segmentation, which is vital for more advanced, efficient, and accurate agricultural monitoring.

Although our approach shows excellent performance in remote sensing crop segmentation, future work may benefit from exploring several potential directions. The extension of MAF-SAM to hyperspectral remote sensing crop segmentation is the focus of the next work. In addition, developing crop-specific, large-scale segmentation models for multispectral remote sensing data remains for future investigation.

References

[1] T. E. Anju and S. Vimala, "A study on digital pathology image segmentation using deep learning methods," in *Proc. 9th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, vol. 1, Mar. 2023, pp. 132–136.

[2] M. Gao, F. Zheng, J. J. Q. Yu, C. Shan, G. Ding, and J. Han, "Deep learning for video object segmentation: A review," *Artif. Intell. Rev.*, vol. 56, no. 1, pp. 457–531, Apr. 2022, doi: 10.1007/s10462-022-10176-7.

[3] P. M. Bhatt et al., "Image-based surface defect detection using deep learning: A review," *J. Comput. Inf. Sci. Eng.*, vol. 21, no. 4, Feb. 2021, Art. no. 040801. [Online]. Available: https://doi.org/10.1115/1.4049535

[4] J. Li, X. Huang, L. Tu, T. Zhang, and L. Wang, "A review of building detection from very high resolution optical remote sensing images," *GISci. Remote Sens.*, vol. 59, no. 1, pp. 1199–1225, Dec. 2022.

[5] B. Wu, J. Meng, Q. Li, N. Yan, X. Du, and M. Zhang, "Remote sensing-based global crop monitoring: Experiences with China's CropWatch system," *Int. J. Digit. Earth*, vol. 7, no. 2, pp. 113–137, Feb. 2014.

[6] M. Aslahishahri et al., "From RGB to NIR: Predicting of near infrared reflectance from visible spectrum aerial images of crops," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1312–1322.

[7] G. Gutman, S. Skakun, and A. Gitelson, "Revisiting the use of red and near-infrared reflectances in vegetation studies and numerical climate models," *Sci. Remote Sens.*, vol. 4, Dec. 2021, Art. no. 100025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666017221000122

[8] A. M. Ali et al., "Crop yield prediction using multi sensors remote sensing (review article)," *Egyptian J. Remote Sens. Space Sci.*, vol. 25, no. 3, pp. 711–716, Dec. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1110982322000527

[9] S. Khanal, J. Fulton, and S. Shearer, "An overview of current and potential applications of thermal remote sensing in precision agriculture," *Comput. Electron. Agricult.*, vol. 139, pp. 22–32, Jun. 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169916310225

[10] E. Omia et al., "Remote sensing in field crop monitoring: A comprehensive review of sensor systems, data analyses and recent advances," *Remote Sens.*, vol. 15, no. 2, p. 354, Jan. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/2/354

[11] N. Pettorelli, "Satellite remote sensing to support agriculture and forestry," in *Satellite Remote Sensing and the Management of Natural Resources*. London, U.K.: Oxford Univ. Press, 2019, pp. 107–119.

[12] K. Berger et al., "Multi-sensor spectral synergies for crop stress detection and monitoring in the optical domain: A review," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113198.

[13] J. M. Pe na-Barragán, M. K. Ngugi, R. E. Plant, and J. Six, "Object-based crop identification using multiple vegetation indices, textural features and crop phenology," *Remote Sens. Environ.*, vol. 115, no. 6, pp. 1301–1316, Jun. 2011. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425711000290

[14] A. A. Gitelson, Y. J. Kaufman, R. Stark, and D. Rundquist, "Novel algorithms for remote estimation of vegetation fraction," *Remote Sens. Environ.*, vol. 80, no. 1, pp. 76–87, Apr. 2002. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425701002899

[15] J. Huang, H. Wang, Q. Dai, and D. Han, "Analysis of NDVI data for crop identification and yield estimation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 11, pp. 4374–4384, Nov. 2014.

[16] D. A. G. Cárdenas, J. A. Valencia, D. F. A. Velásquez, and J. R. P. Gonzalez, "Dynamics of the indices NDVI and GNDVI in a Rice growing in its reproduction phase from multispectral aerial images taken by drones," in *Advances in Information and Communication Technologies for Adapting Agriculture to Climate Change II*, J. C. Corrales, P. Angelov, and J. A. Iglesias, Eds. Cham, Switzerland: Springer, 2019, pp. 106–119. [Online]. Available: https://api.semanticscholar.org/CorpusID:135352972

[17] B. K. Gelder, A. L. Kaleita, and R. M. Cruse, "Estimating mean field residue cover on midwestern soils using satellite imagery," *Agronomy J.*, vol. 101, no. 3, pp. 635–643, May 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:20:41 UTC from IEEE Xplore. Restrictions apply.

[18] C. Ferencz et al., "Crop yield estimation by satellite remote sensing," *Int. J. Remote Sens.*, vol. 25, no. 20, pp. 4113–4149, Oct. 2004.
[19] C. A. da Silva, M. R. Nanni, P. E. Teodoro, and G. F. C. Silva, "Vegetation indices for discrimination of soybean areas: A new approach," *Agronomy J.*, vol. 109, no. 4, pp. 1331–1343, Jul. 2017.
[20] Z. Zhou, S. Li, and Y. Shao, "Crops classification from Sentinel-2A multi-spectral remote sensing images based on convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 5300–5303.
[21] Y. Wu, P. Wu, Y. Wu, H. Yang, and B. Wang, "Remote sensing crop recognition by coupling phenological features and off-center Bayesian deep learning," *Remote Sens.*, vol. 15, no. 3, p. 674, Jan. 2023.
[22] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.
[23] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.
[24] T. Lu, L. Wan, and L. Wang, "Fine crop classification in high resolution remote sensing based on deep learning," *Frontiers Environ. Sci.*, vol. 10, Oct. 2022, Art. no. 991173. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fenvs.2022.991173
[25] Z. Luo, W. Yang, Y. Yuan, R. Gou, and X. Li, "Semantic segmentation of agricultural images: A survey," *Inf. Process. Agricult.*, vol. 11, no. 2, pp. 172–186, Jun. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214317323000112
[26] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.
[27] J. K. Jadhav and R. P. Singh, "Automatic semantic segmentation and classification of remote sensing data for agriculture," *Math. Models Eng.*, vol. 4, no. 2, pp. 112–137, Jun. 2018, doi: 10.21595/mme.2018.19840.
[28] R. M. Rustowicz, R. Cheong, L. Wang, S. Ermon, M. Burke, and D. Lobell, "Semantic segmentation of crop type in Africa: A novel dataset and analysis of deep learning methods," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 75–82.
[29] N. V. S. Natteshan and N. S. Kumar, "Effective SAR image segmentation and classification of crop areas using MRG and CDNN techniques," *Eur. J. Remote Sens.*, vol. 53, no. sup1, pp. 126–140, Jun. 2020, doi: 10.1080/22797254.2020.1727777.
[30] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.
[31] M. A. Mazurowski, H. Dong, H. Gu, J. Yang, N. Konz, and Y. Zhang, "Segment anything model for medical image analysis: An experimental study," *Med. Image Anal.*, vol. 89, Oct. 2023, Art. no. 102918. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361841523001780
[32] P. Shi, J. Qiu, S. M. D. Abaxi, H. Wei, F. P.-W. Lo, and W. Yuan, "Generalist vision foundation models for medical imaging: A case study of segment anything model on zero-shot medical segmentation," *Diagnostics*, vol. 13, no. 11, p. 1947, Jun. 2023. [Online]. Available: https://www.mdpi.com/2075-4418/13/11/1947
[33] W. Ji, J. Li, Q. Bi, T. Liu, W. Li, and L. Cheng, "Segment anything is not always perfect: An investigation of SAM on different real-world applications," 2023, *arXiv:2304.05750*.
[34] A. B. Camiletto, A. Bochicchio, A. Liniger, D. Dai, and A. Gawel, "U-BEV: Height-aware bird's-eye-view segmentation and neural map-based relocalization," 2023, *arXiv:2310.13766*.
[35] M. Zhu et al., "SegPrompt: Boosting open-world segmentation via category-level prompt learning," 2023, *arXiv:2308.06531*.
[36] J. Wang, X. Long, G. Chen, Z. Wu, Z. Chen, and E. Ding, "U-HRNet: Delving into improving semantic representation of high resolution network for dense prediction," 2022, *arXiv:2210.07140*.
[37] C. Ziwen et al., "AutoFocusFormer: Image segmentation off the grid," 2023, *arXiv:2304.12406*.
[38] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin Transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9686686
[39] S. Piramanayagam, W. Schwartzkopf, F. W. Koehler, and E. Saber, "Classification of remote sensed images using random forests and deep learning framework," *Proc. SPIE*, vol. 10004, Oct. 2016, Art. no. 100040L, doi: 10.1117/12.2243169.
[40] L. Wang, J. Wang, X. Zhang, L. Wang, and F. Qin, "Deep segmentation and classification of complex crops using multi-feature satellite imagery," *Comput. Electron. Agricult.*, vol. 200, 2022, Art. no. 107249. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169922005634
[41] L. Wang, J. Wang, Z. Liu, J. Zhu, and F. Qin, "Evaluation of a deep-learning model for multispectral remote sensing of land use and crop classification," *Crop J.*, vol. 10, no. 5, pp. 1435–1451, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214514122000411
[42] C. Xu, M. Gao, J. Yan, Y. Jin, G. Yang, and W. Wu, "MP-Net: An efficient and precise multi-layer pyramid crop classification network for remote sensing images," *Comput. Electron. Agricult.*, vol. 212, Sep. 2023, Art. no. 108065. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169923004532
[43] Z. Du, J. Yang, C. Ou, and T. Zhang, "Smallholder crop area mapped with a semantic segmentation deep learning method," *Remote Sens.*, vol. 11, no. 7, p. 888, Apr. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/7/888
[44] X. Zhao et al., "Use of unmanned aerial vehicle imagery and deep learning UNet to extract Rice lodging," *Sensors*, vol. 19, no. 18, p. 3859, Sep. 2019.
[45] H. Wang, X. Chen, T. Zhang, Z. Xu, and J. Li, "CCTNet: Coupled CNN and transformer network for crop segmentation of remote sensing images," *Remote Sens.*, vol. 14, no. 9, p. 1956, Apr. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/9/1956
[46] G. Chamundeeswari, S. Srinivasan, S. P. Bharathi, P. Priya, G. R. Kannammal, and S. Rajendran, "Optimal deep convolutional neural network based crop classification model on multispectral remote sensing images," *Microprocessors Microsystems*, vol. 94, Oct. 2022, Art. no. 104626. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0141933122001636
[47] Y. Bian, L. Li, and W. Jing, "CACPU-Net: Channel attention U-Net constrained by point features for crop type mapping," *Frontiers Plant Sci.*, vol. 13, 2023, Art. no. 1030595. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fpls.2022.1030595
[48] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," Oct. 2018, *arXiv:1810.04805*.
[49] L. Ouyang et al., "Training language models to follow instructions with human feedback," 2022, *arXiv:2203.02155*.
[50] H. Touvron et al., "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
[51] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in HR remote sensing images," 2023, *arXiv:2309.01429*.
[52] T. Chen et al., "SAM fails to segment anything? SAM-adapter: Adapting SAM in underperformed scenes: Camouflage, shadow, medical image segmentation, and more," 2023, *arXiv:2304.09148*.
[53] J. Liu, Y. Wang, C. Ju, C. Ma, Y. Zhang, and W. Xie, "Annotation-free audio-visual segmentation," 2023, *arXiv:2305.11019*.
[54] S. Liu, J. Chen, B.-G. He, T. Chen, G. Jeon, and W. Wang, "Adapting segment anything model for shield tunnel water leakage segmentation," in *Proc. Workshop Adv. Multimedia Comput. Smart Manuf. Eng.* New York, NY, USA: Association for Computing Machinery, Oct. 2023, pp. 13–18, doi: 10.1145/3606042.3616453.
[55] Z. Zhong, Z. Tang, T. He, H. Fang, and C. Yuan, "Convolution meets loRA: Parameter efficient finetuning for segment anything model," Jan. 2024, *arXiv:2401.17868*.
[56] S. Shankar, L. A. Stearns, and C. J. van der Veen, "Segment anything in glaciology: An initial study implementing the segment anything model (sam)," *Res. Square*, pp. 1–10, 2023. [Online]. Available: https://www.webofscience.com/wos/alldb/full-record/WOS:001108958000001
[57] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11953–11965.
[58] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.
[59] C. Li, H. Farkhoor, R. Liu, and J. Yosinski, "Measuring the intrinsic dimension of objective landscapes," 2018, *arXiv:1804.08838*.
[60] E. J. Hu et al., "LoRa: Low-rank adaptation of large language models," Jun. 2021, *arXiv:2106.09685*.
[61] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," 2015, *arXiv:1508.00092*.
[62] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3322–3325.
[63] C. Zhang et al., "A segmentation model for extracting farmland and woodland from remote sensing image," *Preprints*, Dec. 2017, doi: 10.20944/preprints201712.0192.v1.

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[64] Z. Song, P. Wang, Z. Zhang, S. Yang, and J. Ning, "Recognition of sunflower growth period based on deep learning from UAV remote sensing images," *Precis. Agricult.*, vol. 24, no. 4, pp. 1417–1438, Aug. 2023.
[65] Y. Huang, L. Tang, D. Jing, Z. Li, Y. Tian, and S. Zhou, "Research on crop planting area classification from remote sensing image based on deep learning," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Dec. 2019, pp. 1–4.
[66] C. Yan et al., "HyFormer: Hybrid transformer and CNN for pixel-level multispectral image land cover classification," *Int. J. Environ. Res. Public Health*, vol. 20, no. 4, p. 3059, Feb. 2023. [Online]. Available: https://www.mdpi.com/1660-4601/20/4/3059
[67] J. Xiang, J. Liu, D. Chen, Q. Xiong, and C. Deng, "CTFuseNet: A multi-scale CNN-transformer feature fused network for crop type segmentation on UAV remote sensing imagery," *Remote Sens.*, vol. 15, no. 4, p. 1151, Feb. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/4/1151
[68] X. Ma, X. Zhang, and M.-O. Pun, "A crossmodal multiscale fusion network for semantic segmentation of remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3463–3474, 2022.

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